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ORIGINAL PAPER

Prevalence and molecular identification of *Giardia duodenalis* in different water sources in Baghdad City

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Abstract

Giardia duodenalis is a major contributor to the global burden of waterborne diseases, with drinking and recreational water sources vulnerable to fecal contamination from human and animal waste. Between November 2021 and August 2022, a total of 400 source water samples were collected from eight different supply systems in Baghdad. Samples were analyzed using both traditional microscopy and nested polymerase chain reaction (nPCR) to detect the *Giardia duodenalis* cysts. The results revealed that 7% of samples were positively identified by microscopy while nPCR demonstrated higher prevalence 12.5% ($P \leq 0.01$) confirming the greater sensitivity of molecular techniques. Lugol's iodine staining performed better than wet mount in microscopic detection. The highest contamination was observed in July with nPCR identifying up to 39.39% positivity, while no cyst were detected during the winter months. Geographically, the eastern part of Baghdad, especially Rashidiya, had the highest prevalence (16%) in contrast to the western side (8%). Physicochemical parameters were also evaluated and showed strong correlations with *Giardia* prevalence. Higher contamination was associated with elevated total dissolved solids (up to 534 ppm), nitrate (up to 15 mg/L), and turbidity (up to 8.4 NTU), particularly in sites such as Rashidiya and Al-Nahrawan. pH remained neutral (about 7.5), while temperature ranged from 22.8 °C to 25.4 °C. Rainfall and humidity patterns were relatively stable across sites. Furthermore, electrical conductivity (704–804 µS/cm) and total dissolved solids exceeded, in some cases, the USA EPA freshwater standard (500 ppm), reflecting localized anthropogenic impacts. This study highlights the role of environmental factors - such as temperature, nitrate, total dissolved solids, and turbidity - in shaping the prevalence of *G. duodenalis* in Baghdad's water sources. It underscores the need for integrated monitoring strategies that combine molecular diagnostics with water quality surveillance to safeguard public health. To the best of our knowledge, this is the first comprehensive molecular survey of *Giardia duodenalis* across multiple water source systems in Baghdad. The study's novelty lies in its large-scale sampling effort, integration of advanced molecular diagnostics (nPCR) with conventional microscopy, and the inclusion of detailed physicochemical and seasonal analyses to better understand environmental risk factors in an underreported Middle Eastern urban setting.

Keywords Giardia duodenalis · Baghdad · Nested PCR · Turbidity · Nitrate · Seasonal variation · Waterborne protozoa

Introduction

Giardia duodenalis (also known as *G. lamblia*) is a flagellated protozoan responsible for giardiasis, a globally widespread intestinal disease primarily transmitted through water contaminated with human or animal feces (Feng and Xiao 2011). In developed countries, prevalence rates typically range from 2 to 5%, while in developing regions, they may reach 20–30% (Dunn and Juergens 2024).

In Iraq, giardiasis is considered an endemic disease, with infection rates varying depending on the geographical location and diagnostic method used. For example, in Babylon province, a study involving children and calves revealed

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that infection was prevalent in both hosts, and the zoonotic transmission was confirmed through molecular analysis of the tpi gene (Alseady et al. 2023). In Thi-Qar province, another study conducted in Al-Rifai district showed that the infection rate detected via PCR was 57.3% among diarrheic patients, with the highest prevalence observed in the 1–10 year age group, particularly in rural areas (Al-Aboody et al. 2020).

The cysts of *G. duodenalis* are notable for their high resistance and persistence, as they can withstand conventional disinfection methods such as chlorination, allowing them to survive in water sources even after treatment (Wang et al. 2023). Environmental factors such as turbidity, total dissolved solids (TDS), nitrate concentration, and seasonal temperature fluctuations have been associated with increased presence of cysts in various aquatic environments (Bada et al. 2024; Olukosi et al. 2016; Hatam-Nahavandi et al. 2025).

Despite the public health significance of this parasite, comprehensive data on the prevalence of *G. duodenalis* in Baghdad remain scarce. Most previous studies relied on traditional microscopic examination, which lacks the sensitivity needed to detect low-level contamination and does not provide genotypic characterization of the parasite (Fradette et al. 2022; Hatam-Nahavandi et al. 2024) To date, there have been no published studies in Baghdad combining molecular diagnostics (such as nested PCR), seasonal and spatial sampling, and water quality analysis to investigate environmental drivers of giardiasis.

Water is one of the most essential natural resources supporting all forms of plant and animal life (Abida and Hankrishna 2008). It possesses unique chemical properties due to its polarity and hydrogen bonding capability, allowing it to dissolve, absorb, adhere to, and suspend a wide variety of compounds (WHO 2004). As a result, water in its natural state is never pure, often acquiring pollutants from its surrounding environment, including those derived from humans, animals, and other biological activities (Abida and Hankrishna 2008).

Access to high-quality drinking water is critical to human health, as survival largely depends on its availability (Lamikanra 1999). Ensuring safe drinking water for both rural and urban populations is essential in preventing various health risks (Lomniezi et al. 2007).

Globally, many parasitic diseases are linked to contaminated water sources, including giardiasis. Infection typically occurs when individuals ingest or come into contact with contaminated water. Therefore, proactive measures to monitor water quality are necessary, along with targeted interventions to prevent further degradation (Sha-Amu et al. 2024).

Relying solely on traditional methods to monitor water quality is no longer sufficient (Ritchie et al. 2003). Integrating traditional techniques with remote sensing technologies has proven effective in water quality monitoring by providing broader spatial and temporal coverage (Dlamini et al. 2016).

In areas with low population density, surface water bodies such as streams, lakes, and ponds are particularly vulnerable to fecal contamination due to the absence or inefficiency of sewage infrastructure. As rainwater or surface runoff moves across land, it collects various contaminants, including microorganisms, organic matter, and minerals. The high nutrient content of such water creates favorable conditions for microbial proliferation, posing significant risks to both environmental and public health.

The novelty of this study lies in its integrated and context-specific approach. It is the first research conducted in Baghdad that combines traditional parasitological diagnostics (direct wet mount and Lugol's iodine stain) with molecular detection using nested PCR targeting the tpi gene. Furthermore, the study uniquely examines spatial and seasonal variations and correlates *Giardia* prevalence with key physicochemical indicators of water quality such as turbidity, TDS, and nitrate levels. This multidisciplinary framework provides new insights into the environmental drivers of *G. duodenalis* contamination in an urban setting with complex water infrastructure and limited prior molecular surveillance.

Accordingly, this study aims to:

1. Determine the prevalence of *G. duodenalis* in various water systems in Baghdad using microscopic examination and nested PCR,
2. Evaluate seasonal and spatial variations, and.
3. Investigate the relationship between parasite prevalence and certain chemical indicators of drinking water such as turbidity, TDS, and nitrate.

Despite the global awareness of *Giardia duodenalis* as a waterborne pathogen, there remains a significant gap in localized studies, particularly in regions like Baghdad. The city is characterized by a rapidly growing population, a mix of urban and peri-urban areas, and reliance on multiple and often unregulated water sources, including rivers, open drainage systems, and groundwater tanks. These factors increase the risk of water contamination with human and animal fecal matter, especially in the absence of consistent water quality monitoring. To date, there is a lack of comprehensive molecular-based investigations into the prevalence and distribution of *G. duodenalis* in Baghdad's water systems. Therefore, this study aims to address this gap by evaluating the presence of *G. duodenalis* cysts in various water sources across Baghdad using both traditional and molecular diagnostic methods, with the goal of contributing

to better public health awareness and improved water management policies in the region.

Given the limited molecular investigations conducted in Baghdad and the increasing reliance on multiple, potentially contaminated water sources, this study provides essential insight into the public health burden and environmental dynamics of *Giardia duodenalis* in a region with complex urban infrastructure.

Materials and methods

Samples collection

Water samples were collected from eight distinct locations within the Baghdad governorate to represent a variety of environmental and anthropogenic influences, including urban, peri-urban, rural, and surface water sources. The sampling sites included rivers, irrigation canals, tap water storage tanks, and surface pools, covering both residential and agricultural zones. The geographic coordinates (latitude and longitude) of each sampling point were determined using Google Maps and are presented in Table 1 to ensure spatial reproducibility.

Samples of surface and tap water were collected in sterile polyethylene bottles (2–3 L capacity) from the selected sites. The collected water samples were transported in cooled insulated containers (4 ± 2 °C) to the laboratory and processed within 24 h to preserve microbial and protozoan integrity. These sampling locations were chosen to evaluate the spatial distribution and prevalence of *Giardia duodenalis* in diverse aquatic environments-ranging from municipal water supplies and household storage tanks to rivers, agricultural drainage canals, and stagnant surface water bodies-that may be impacted by human and animal fecal contamination.

Table 1 Geographic coordinates and characteristics of the water sampling sites in Baghdad

Site No.	Location Description	Type	Latitude (*N*)	Longitude (E)
1	Abu Ghraib drainage	Agricultural drainage	33.3092°	44.1093°
2	Sabaa Al-Bor drainage	Urban-rural interface	33.4586°	44.1958°
3	Al-Rashidiya drainage	Suburban drainage	33.4317°	44.4783°
4	Al-Kadhimiya drainage	Urban drainage	33.3772°	44.3556°
5	Al-Nahrawan & Al-Madain drainage	Agricultural drainage	33.1928°	44.6625°
6	Puddles and streams (various)	Stagnant surface water	Various	Various
7	Tigris River	Major surface water	33.3128°	44.3615°
8	Household tanks (various neighborhoods)	Stored water supply	Various	Various

Description of sampling sites

Eight distinct water source sites were selected across Baghdad, Iraq, to assess the prevalence of *Giardia duodenalis* in diverse environmental contexts. The selection aimed to capture a wide range of hydrological, geographical, and anthropogenic influences. Table 1 summarizes the key characteristics of these sites, including their type, exact coordinates (when available), and contextual setting.

1. Abu Ghraib Drainage (Site 1) -This site represents agricultural drainage water, located in a heavily irrigated farming area in western Baghdad (33.3092°N, 44.1093°E). It receives runoff from surrounding agricultural lands, potentially contaminated with animal feces and fertilizers.
2. Sabaa Al-Bor Drainage (Site 2) - Situated at the urban-rural interface (33.4586°N, 44.1958°E), this site receives mixed discharge from both residential settlements and adjacent farmlands, making it a transitional water body with complex contamination profiles.
3. Al-Rashidiya Drainage (Site 3) – A suburban drainage site located in northeastern Baghdad (33.4317°N, 44.4783°E), characterized by a mix of domestic wastewater inputs and limited agricultural runoff.
4. Al-Kadhimiya Drainage (Site 4) – Located in a densely populated urban area (33.3772°N, 44.3556°E), this drainage system collects untreated domestic wastewater, posing a high risk of protozoan contamination due to poor sanitation infrastructure.
5. Al-Nahrawan & Al-Madain Drainage (Site 5) – This site represents agricultural drainage in the southeastern zone of Baghdad (33.1928°N, 44.6625°E). It is primarily influenced by irrigation return flows and livestock activity.
6. Puddles and Streams (Site 6) – These are stagnant surface water bodies located in various urban and peri-urban neighborhoods. Their coordinates vary, but they typically accumulate rainwater, untreated effluent, and urban runoff, becoming breeding grounds for microbial pathogens due to stagnant conditions.
7. Tigris River (Site 7) – As the main surface water body in Baghdad (33.3128°N, 44.3615°E), the Tigris River was sampled to serve as a baseline reference. Despite its high flow, it is susceptible to upstream contamination from untreated wastewater and agricultural runoff.
8. Household Tanks (Site 8) – These are stored water sources located in various residential neighborhoods across Baghdad. While the coordinates vary, these tanks

often store water for extended periods, which can affect water quality, especially when hygiene and maintenance practices are inadequate.

A total of 400 water source samples were collected from 8 different water source systems and supplies in the north, west, east and south Baghdad city, including Abu Ghraib drainage system, Sabaa Al-Bor drainage system, Rashidiya drainage system, Kadhimiya drainage system, Al-Nahrawan and Al-Madain drainage system, puddles and streams, Tigris river and home tanks water supplies. 50 samples per station, approximately 2–3 L in sterile bottles, were collected during the period from November 2021 and August 2022 for the detection of *Giardia* cysts using traditional and nested PCR methods.

Methodological relevance and transferability

The methodology employed in this study, combining traditional diagnostic techniques and nested PCR, proved effective in detecting *Giardia duodenalis* in diverse urban water sources. Given its reliability and adaptability, this approach can be applied to other urban settings with varying levels of water source contamination to support public health surveillance and water safety assessments.

Ethical approval statement

As the principal investigator is a full-time employee at the Ministry of Water Resources, the sample collection was conducted as part of her official duties. A formal No Objection Certificate (NOC) was issued by the Ministry after reviewing the research objectives and confirming that the study aligns with institutional priorities. Therefore, no additional ethical approval was required. All procedures for sample collection and handling adhered strictly to governmental regulations and scientific standards.

Sample size justification

The decision to collect 50 samples per water source system (totaling 400 samples) was based on logistical feasibility and the need for representative data across urban water infrastructures. This approach aligns with methodologies employed in comparable Mediterranean and European studies. For instance, Plutzer et al. (2007) examined 76 raw and drinking water samples in Hungary—supplemented by 157 additional samples from the Danube and riverbank-filtered sources—and reported *Giardia* cysts in 48.4% of raw and 26.7% of treated samples, supporting the utility of medium-to-large sampling frames in protozoan monitoring (Plutzer et al. 2007). Similarly, Ben Ayed et al. (2012) analyzed 232 samples collected from 18 wastewater treatment plants across Tunisia using PCR-based assays, demonstrating the effectiveness of moderate sampling scales in detecting *Giardia*, *Cryptosporidium*, and other enteric protozoa across geographically diverse sites. In Spain, Ramo et al., (2016) surveyed water from 20 treatment plants across the Aragón region and identified protozoan contamination in over 70% of sites, underscoring how targeted, moderately scaled sampling strategies can effectively reveal contamination patterns. Together, these studies reinforce the epidemiological validity of using 400 samples, which were sufficient in this study to detect statistically significant differences ($P \leq 0.01$) and provide meaningful public health insights.

Samples preparation

Water samples after collection were sent to a private "Scientific Source Laboratory". Each sample was divided and placed in test tubes and centrifuged at 3000 rpm for 5–10 min. Sediment with rare solution was collected and approximately 10 ml of sediment was divided into two parts. The first one (about 2 ml) for direct examination by direct wet mount and Lugol's iodine stain smears (Soulsby 1982).

Microscopic examination

In this study, microscopic examinations were performed directly by the primary researcher, who also conducted the sample collection. As such, the identity and source of each sample were known during analysis, which introduces a recognized limitation in terms of blinding. To mitigate this, all examinations were conducted in collaboration with experienced laboratory technicians, and strict adherence to standard parasitological diagnostic criteria was maintained throughout the analysis.

Sample processing

Direct wet mount and Lugol's iodine stain are commonly used as preliminary screening tools for morphological identification of *Giardia* cysts due to their simplicity and cost-effectiveness in routine parasitological diagnostics (Garcia 2007). However, these microscopic methods have limited sensitivity, especially when cyst counts are low or samples are turbid (Feng and Xiao 2011). Therefore, molecular methods such as nested PCR (nPCR), targeting the triosephosphate isomerase (tpi) gene, have been widely adopted to improve detection sensitivity and enable genetic confirmation of *Giardia duodenalis* (Sulaiman et al. 2003; Caccio and Ryan 2008).

Briefly, direct wet mount and Lugol's iodine staining provide visual confirmation of *Giardia* cysts, while nPCR

enhances diagnostic accuracy by detecting specific DNA sequences, allowing identification even in low-level contamination or degraded samples (Xiao and Fayer 2008). This combined approach ensures both morphological and molecular verification, offering a more comprehensive assessment of *Giardia* contamination (Ryan and Caccio 2013).

Physicochemical water quality analysis

In response to growing evidence regarding the impact of environmental factors on the survival and detection of *Giardia duodenalis*, supplementary analyses were conducted on retained water samples to evaluate key physicochemical parameters. These included turbidity, pH, temperature, electrical conductivity (EC), total dissolved solids (TDS), and nitrate concentration. Measurements were carried out under controlled conditions using calibrated instruments, and the resulting data were incorporated into this study to support the interpretation of parasitological findings.

This multifactorial approach was based on previous regional and international studies that demonstrated a correlation between water quality indicators and protozoan contamination (Wang et al. 2023; Bada et al. 2024; Olukosi et al. 2016). Incorporating these variables enriches the ecological context of *Giardia* distribution across different water sources, thereby enhancing the public health relevance of the findings.

DNA extraction and nested PCR amplification

Genomic DNA was extracted from *Giardia duodenalis* cysts using a method based on multiple freeze–thaw cycles to ensure the disruption of the cyst wall, followed by purification using the G-spin™ Total DNA Extraction Kit (Intron Biotechnology, Cat. No. 17045, Korea). A study by Adamska et al. (2010) demonstrated that freezing in liquid nitrogen followed by boiling, in combination with reliable extraction kits, significantly enhanced DNA extraction efficiency and improved the sensitivity of molecular detection using nested PCR. Similarly, Sepahvand et al. (2017) confirmed the effectiveness of repeated freeze–thaw cycles in facilitating DNA release from *Giardia* cysts.

The concentration and purity of the extracted DNA were measured using the Quantus™ Fluorometer and the QuantiFluor dsDNA system, ensuring DNA quality suitable for PCR applications.

For the nested PCR assay, the triose phosphate isomerase (tpi) gene was selected as the target. Primers were adopted from the widely referenced protocol by Sulaiman et al. (2003), which is commonly used for the molecular detection and genotyping of *Giardia*. This approach offers higher sensitivity and superior genotypic characterization compared to conventional microscopic methods.

Finally, DNA amplification was performed using the Techne PrimeG Gradient Thermal Cycler under optimized thermal conditions, and the PCR products were visualized on a 2% agarose gel for verification and isolation.

The primer sets used were those previously published by Sulaiman et al. (2003), as follows:

- First-round primers:
- AL3543-F: 5′-AAATIATGCCTGCTCGTCG-3′.
- AL3546-R: 5′-CAAACCTTITCCGCAAACC-3′.
- Second-round primers:
- AL3544-F: 5′-CCCTTCATCGGGGTAACTT-3′.
- AL3545-R: 5′-GTGGCCACCACICCCGTGCC-3′.

The PCR reactions were carried out using a Techne PrimeG Gradient Thermal Cycler (Bibby Scientific Ltd., UK). Amplification conditions were optimized as follows:

- Initial denaturation at 94 °C for 3 min (1 cycle).
- Denaturation at 94 °C for 45 s.
- Annealing at 60 °C for 30 s.
- Extension at 72 °C for 1 min (35 cycles).
- Final extension at 72 °C for 7 min.

Gel electrophoresis was performed on 2% agarose gels to visualize the amplified products. The electrophoresis was run at 70 V and 65 mA for 1 h using a Consort EV231 Power Supply (Belgium). DNA bands were observed under ultraviolet light using a UVP Trans-illuminator Model TFL-20 M.

These methods, particularly the combination of nested PCR and microscopy, have been successfully applied in previous studies to detect and genotype *Giardia*, and are suitable for use in diverse epidemiological and environmental settings.

Nested PCR primer design and validation

The primer sequences used in this study were selected to amplify a fragment of the *Giardia duodenalis* triose phosphate isomerase (TPI) gene, based on the established protocol by Sulaiman et al. (2003). The first-round PCR utilized primers AL3543 (forward: 5′-AAATIATGCCTGCTCGTCG-3′) and AL3546 (reverse: 5′-CAAACCTTITCCGCAAACC-3′), while the second-round (nested) PCR employed primers AL3544 (forward: 5′-CCCTTCATCGGGGTAACTT-3′) and AL3545 (reverse: 5′-GTGGCCACCACICCCGTGCC-3′).

The PCR conditions-annealing temperature (60 °C), number of cycles (35), and primer concentration (10 pmol/µL)-were directly adopted from Sulaiman et al.,'s protocol

without further modification. This decision was based on the robust performance of the protocol in earlier research and its wide adoption in subsequent studies (e.g., Adamska et al. 2010; Jerez Puebla et al. 2015), which reported consistent amplification of *Giardia* assemblages from environmental matrices. Thus, no additional primer optimization was necessary for this study.

Primers used

The nested PCR employed AL3543F and AL3546R primers targeting a fragment of the triose phosphate isomerase (Tpi) gene, a commonly used genetic marker for *Giardia duodenalis*. These primers were selected based on their high sensitivity and specificity, as reported by Sulaiman et al. (2003) and Read et al. (2002). The Tpi gene offers reliable genotyping capabilities, especially for distinguishing between zoonotic assemblages A and B, which are of public health importance. This primer set has been successfully applied in environmental surveillance and clinical diagnostics due to its ability to detect low cyst concentrations in complex matrices.

Optimal nPCR conditions

The optimal nPCR conditions are initial denaturation (94 °C for 3 min. one cycle), denatureation-2 (94 °C, 45s), annealing (60 °C, 30 s), extension-1 (72 °C for 1 min) for 35 cycle and extention-2 (72°C, 7 min., one cycle) for both nPCR runs.

The primers preparation

Lyophilized primers were dissolved in ddH_2O prepare a stock solution with a final concentration of 100 pmol/µl and store it at -20 °C. To make a working primer suspension at a concentration of 10 pmol/µl, mix 10 µl of the stock solution with 90 µl of ddH_2O, yielding a total volume of 100 µl.

Nanodrop calibration

The Quantus™ Fluorometer was used with the QuantiFluor® Dye Systems to estimate DNA concentration and purity.

DNA concentration and purity were measured using the Quantus™ Fluorometer (Promega) with QuantiFluor® dye systems, and further verified by NanoDrop spectrophotometry. The A260/A280 ratios ranged from 1.49 to 1.97, with the majority falling within the acceptable range of 1.7–2.0, indicating sufficient purity for molecular analysis.

Gel electrophoresis

To prepare the agarose gel, 2 g Agarose powder was dissolved in 100 ml of 1X TBE in a 100 ml flask. The mixture was heated in a hot block until it became clear. Once the agarose solution was prepared, it was allowed to cool to approximately 50–55 °C, with occasional swirling of the flask to ensure even cooling. Next, 3 µl of red stain was added to the warm gel. The ends of the casting tray were sealed with two layers of tape, and the combs were positioned in the tray. The melted agarose solution was then poured into the casting tray and allowed to solidify at room temperature. After solidification, the comb was carefully removed, and the tape was taken off. The gel was then placed in the electrophoresis chamber, which was filled with 1X TBE buffer. DNA samples (5 µl) were mixed with 3 µl of DNA loading buffer and loaded into the wells of the agarose gel. Agarose gel electrophoresis was performed. At 70 V, 65 mA for 1 h and DNA was observed under a UV trans-illuminator.

Measurement of physicochemical parameters

Physicochemical parameters were measured at each sampling site and during all sampling seasons to evaluate environmental conditions potentially influencing the presence of *Giardia duodenalis.* The parameters included pH, temperature (°C), turbidity (NTU), electrical conductivity (EC, µS/cm), total dissolved solids (TDS, mg/L), nitrate concentration (NO_3^-, mg/L), rainfall (mm), and relative humidity (%).

Measurements of pH, EC, TDS, and temperature were performed in situ using a portable multi-parameter meter (HANNA HI98129, Hanna Instruments, Italy) following APHA Standard Method 2510 B for EC and Method 2550 B for temperature. Turbidity was measured using a portable turbidity meter (HACH 2100Q, Hach Company, USA) in accordance with APHA Method 2130 B. Nitrate concentrations were analyzed in the laboratory using a UV-Visible spectrophotometer (Shimadzu UV-1800, Shimadzu Corporation, Japan) following the cadmium reduction method (APHA Method 4500-$NO3^-$ E). Rainfall and relative humidity data were obtained from the Iraqi Meteorological Organization records corresponding to each sampling location and time. All procedures were conducted according to the Standard Methods for the Examination of Water and Wastewater (APHA 2017).

Molecular detection of *Giardia duodenalis* by nested PCR

Nested PCR (nPCR) was employed for the molecular detection of *Giardia duodenalis* in environmental water samples due to its superior sensitivity and specificity, particularly

in samples with low concentrations of target DNA and the potential presence of PCR inhibitors. The method targeted the triosephosphate isomerase (TPI) gene, following the protocol established by Sulaiman et al. (2003). To minimize the risk of cross-contamination-a known limitation of nPCR-strict laboratory practices were followed. These included the use of negative controls in each run, physical separation between pre- and post-PCR work areas, and adherence to clean-room protocols with dedicated pipettes and filtered.

Laboratory facilities and author's role

All laboratory procedures-including sample processing, DNA extraction, and nested PCR assays-were carried out entirely by the principal investigator (the author) in a privately operated, non-commercial molecular biology laboratory located in Baghdad. This laboratory is not affiliated with any commercial enterprise. Rather, it functions as a scientific facility under academic supervision and is frequently utilized by postgraduate students from various universities in Iraq.

The laboratory is well-equipped with essential instruments such as thermal cyclers, gel electrophoresis units, fluorimeters, and UV transilluminators, enabling comprehensive molecular diagnostics. Moreover, the facility provides free theoretical and practical training to postgraduate students, contributing to capacity building in molecular biology and public health research.

All experimental procedures were personally performed by the author to ensure data integrity, reproducibility, and methodological consistency.

Statistical analysis

The Statistical Analysis System (SAS) software was used to assess the effects of different factors on the study parameters, and the Chi-square (χ^2) test was applied to determine statistical significance at probability levels of 0.05 and 0.01.

In addition, descriptive statistics (mean ± standard deviation) were used to summarize the physicochemical water quality parameters measured across different sampling sites and seasons. Furthermore, Pearson correlation analysis was conducted using SPSS software (version 25) to evaluate the strength and direction of linear relationships between continuous environmental variables (including temperature, pH, turbidity, TDS, and nitrate) and the prevalence of Giardia duodenalis. Results were considered statistically significant at $p < 0.05$.

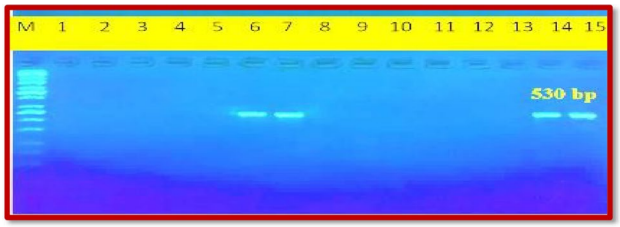

Fig. 1 Electrophoresis on agarose gel (2%). M: 100 bp DNA ladder, lanes 6, 7, 14 and 15 show positive results of the second run of nPCR at 530 bp, consistent with the expected product size for the *Giardia* TPI gene fragment

Results

Identification of *Giardia duodenalis*

Figure 1 illustrates the outcome of the second round of nested PCR amplification targeting the triose phosphate isomerase (TPI) gene of *Giardia duodenalis*. The electrophoresis was performed on a 2% agarose gel stained with ethidium bromide.

- Lane M represents the 100 bp molecular DNA ladder, which serves as a reference to determine the size of the PCR products.
- Lanes 6, 7, 14, and 15 show distinct amplification bands at approximately 530 base pairs, consistent with the expected product size for the *Giardia* TPI gene fragment.
- The presence of these bands confirms successful amplification and the presence of *Giardia* DNA in these water samples.
- The absence of bands in other lanes indicates negative samples or concentrations below the detection limit.

This figure validates the use of nPCR as a sensitive method for detecting low levels of protozoan contamination in environmental samples.

Figure 2 shows microscopic images of *Giardia* cysts observed in water samples using two conventional staining methods:

- Panel A depicts a cyst visualized by the direct wet mount technique. The cyst morphology is faint, and internal structures are less distinguishable due to the lack of contrast.
- Panel B shows a cyst stained with Lugol's iodine, which improves contrast and highlights key internal structures such as nuclei, making identification more reliable.
- Both images were captured under a light microscope at 40× magnification.

The comparison between the two panels demonstrates the enhanced visibility and diagnostic clarity provided by

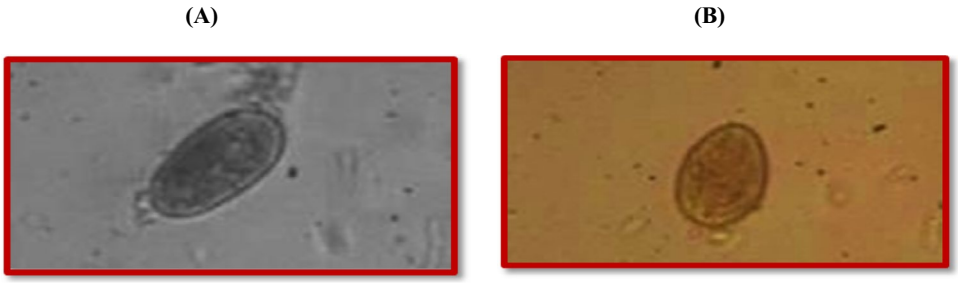

Fig. 2 Cysts of *Giardia* by direct wet mount (**A**) and Lugol's iodine stain (**B**) smears. Both images were captured under a light microscope at 40× magnification

Table 2 Monthly distribution of *Giardia* cysts detected in various water sources using both microscopic examination (wet mounts and smears) and nested PCR (nPCR) techniques

Months	No. of samples examined	Direct wet mount smears		Lugol's Iodine smears		nPCR	
		Positive	(%)	Positive	(%)	Positive	(%)
November	36	0	0	1	2.77	1	2.77
December	34	0	0	0	0	0	0
January	41	0	0	0	0	0	0
February	45	0	0	0	0	0	0
March	41	0	0	2	4.87	2	4.87
April	43	2	4.65	2	4.65	4	9.3
May	41	2	4.87	4	9.75	6	14.63
June	41	3	7.31	7	17.05	10	24.39
July	33	5	15.15	8	24.24	13	39.39
August	45	0	0	4	8.88	14	31.11
Total	400	12	3	28	7	50	12.5
χ2	27.7			32.1		46.8	
P≤			0.001*		0.0002*		0.0001*

Lugol's iodine staining, which aligns with findings that it offers superior detection over unstained wet mounts.

Table 2 summarizes the monthly distribution of *Giardia* cysts detected in various water sources using both microscopic examination (wet mounts and smears) and nested PCR (nPCR) techniques.

Out of 400 water samples collected across ten months, the nPCR method demonstrated a higher overall detection rate (12.5%) compared to traditional microscopy (3%) and direct smears (also 3%), highlighting the superior sensitivity of molecular methods in detecting low levels of parasitic DNA.

Notably, no positive cases were detected from November to February by any method, likely due to lower temperatures and reduced biological activity in water bodies. The first emergence of positive cases began in March, with a noticeable increase in prevalence during the warmer months, peaking in July and August. Specifically, July showed the highest total detection rate (39.39%), with 5 positive cases by smear, 8 by nPCR, and 13 total positives.

The chi-square (χ^2) values for all three detection methods (27.7 for smears, 32.1 for microscopy, and 46.8 for nPCR) were statistically significant ($P \leq 0.001$), indicating a meaningful correlation between month and detection rate. This trend suggests a seasonal pattern, with *Giardia* prevalence increasing during late spring and summer, possibly due to elevated temperatures, increased human activity near water sources, and favorable conditions for cyst survival.

This pattern aligns with findings from a study in the Al-Jinderiyah region of Latakia, Syria, which reported elevated concentrations of *Giardia* cysts during the rainy season (autumn and winter), compared to lower levels in the summer months (Ibrahim et al. 2019). Conversely, a study from Quebec, Canada, documented peak concentrations of both *Giardia* and *Cryptosporidium* during winter and spring. These peaks were associated with snowmelt and seasonal flooding, which increased turbidity and facilitated the transport of cysts into surface waters (Fradette et al., 2022).

Accordingly, the relationship between environmental patterns and seasonality appears well established: in warm or humid regions, *Giardia* prevalence tends to increase during the summer or rainy seasons, whereas in colder climates, peaks are often observed in winter due to hydrological changes like snowmelt. These findings emphasize the importance of seasonally adaptive water quality monitoring and enhanced treatment strategies during periods of hydrological variability.

Table 3 displays the regional distribution of *Giardia* cysts detected in different water sources using three diagnostic approaches: direct wet mount smear, Lugol's iodine smear, and nested PCR (nPCR). A total of 400 samples were

Table 3 Regional distribution of *Giardia* cysts detected in different water sources using three diagnostic approaches: direct wet Mount smear, lugol's iodine smear, and nested PCR (nPCR)

Sources and regions	Number	Direct wet mount smear		Lugol's iodine smears		PCR	
		Positive	Percentage (%)	Positive	Percentage (%)	Positive	Percentage (%)
Kadhimiya	50	1	2	2	4	2	4
Sabaa Al-Bor	50	0	0	2	4	2	4
Rashidiya	50	6	12	8	16	14	28
Nahrawan	50	3	6	8	16	11	22
Abu Ghraib	50	0	0	2	4	7	14
Tigris river	50	0	0	2	4	7	14
Tank water	50	0	0	1	2	1	2
Puddles	50	2	4	3	6	6	12
Total	400	12	3	28	7	50	12.5
χ2	--	21.9	0.0025**	17.2	0.016*	26.9	0.0003***

collected evenly from eight different regions of Baghdad, with 50 samples per region.

The highest prevalence was recorded in Rashidiya, with detection rates reaching 12% by wet mount, 16% by iodine smear, and 28% by nPCR, indicating a significantly higher burden of contamination in this area. This may reflect poor sanitation infrastructure, agricultural runoff, or untreated wastewater exposure.

Other regions with relatively high detection rates include Al- Nahrawan (22% by nPCR) and Abu Ghraib and Tigris River sites (each 14% by nPCR), suggesting widespread contamination in both peripheral urban and river-based sources. In contrast, Sabaa AL-Bor, Kadhimiya, and Tank water sources showed lower levels of contamination, particularly in traditional microscopy, though nPCR still detected positive cases, confirming its superior sensitivity.

The statistical analysis using chi-square (χ^2) tests revealed significant differences in *Giardia* prevalence among regions across all methods:

- Lugol's iodine smear: $\chi^2 = 21.9$, $P = 0.0025$.
- Wet mount smear: $\chi^2 = 17.2$, $P = 0.016$.
- nPCR: $\chi^2 = 26.9$, $P = 0.0003$.

These significant values highlight that geographical location plays a crucial role in waterborne *Giardia* contamination, and emphasize the necessity for region-specific public health interventions and targeted water treatment strategies.

These results are consistent with both local and international studies:

- In southern Iraq (Thi-Qar), the prevalence of *G. lamblia* reached 23.7% among children ($n = 396$), strongly associated with untreated water, rural living conditions, and hot seasons ($p < 0.05$) (Hussain et al. 2010).
- In the Han River, Seoul, South Korea, *Giardia* cysts were found in 35% of raw water samples, with an average concentration of 3.74 cysts/10 L in winter compared to 0.8–1.08 cysts/10 L in other seasons, highlighting the impact of winter weather on cyst occurrence (Lee et al. 2011).
- In Galicia, Spain, *Giardia* was detected in 36.5% of drinking water samples and 60.7% of recreational water samples, with cyst viability ranging from 90 to 95%, underscoring the challenges in water disinfection and environmental monitoring (Castro-Hermida et al. 2010).

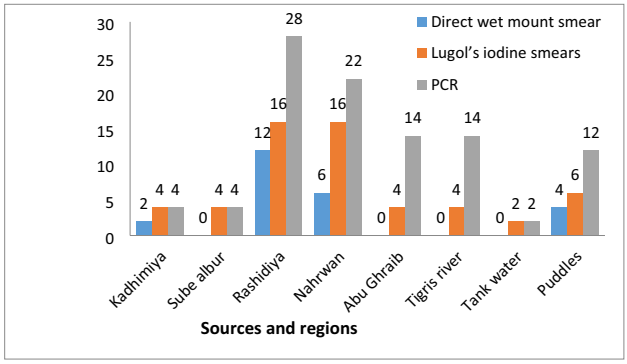

Fig. 3 Distribution of Giardia cyst in water sources according to regions, according to three diagnostic approaches: direct wet mount smear, Lugol's iodine smear, and nested PCR (nPCR)

Figure 3 illustrates the regional distribution of *Giardia* cysts detected in various water sources across different districts of Baghdad using both microscopic examination and nested PCR (nPCR) techniques.

The figure clearly demonstrates that nPCR consistently identified more positive samples than microscopy in all regions, confirming its superior sensitivity in detecting *Giardia duodenalis*. Among the surveyed locations, Rashidiya recorded the highest prevalence, with *Giardia* detected in up to 28% of samples by nPCR, followed by Al-Nahrawan (22%), Abu Ghraib (14%), and Tigris River (14%).

In contrast, areas such as Tank water, Kadhimiya, and Sabaa Al-Bor showed the lowest detection rates, especially with microscopy, suggesting either better water quality or limitations in traditional diagnostic methods.

This is consistent with the findings of Feng and Xiao (2011), who emphasized the higher sensitivity of molecular methods. Regional contamination rates are influenced by environmental factors and infrastructure, as reported in international studies (Karanis et al. 2007; Wang et al. 2023). Furthermore, the review by Fradette et al. (2022) supports the effectiveness of nested PCR (nPCR) in detecting protozoan contamination in environmental water samples.

Table 4 presents the regional distribution of *Giardia duodenalis* infections based on nested PCR (nPCR) results, grouped into three main geographical sectors of Baghdad: north, east, and west.

Out of a total of 400 samples, the highest infection rate was recorded in the eastern region of Baghdad, which includes Al-Nahrawan, Puddles, and Tigris River, where 24 samples (16%) tested positive. This was followed by the northern region (Kadhimiya, Sabaa Al-Bor, and Rashidiya) with 18 positive cases (12%), while the western region (Abu Ghraib and tank water sources) had the lowest prevalence, with only 8 cases (8%).

These findings indicate a clear spatial variation in *Giardia* contamination levels, possibly reflecting differences in wastewater management, population density, industrial/agricultural runoff, or water infrastructure between regions.

Local studies indicate significant variation in the prevalence of *Giardia duodenalis* infections across Iraq. High infection rates have been recorded among children in governorates such as Thi-Qar (23.7%), with a clear association with hot seasons, untreated water, and rural settings (Hussein 2010). In Duhok, earlier studies reported a prevalence of 38.5%, whereas a more recent study conducted between 2022 and 2023 found a reduced rate of 4.9% (Ismael et al. 2024). In Baghdad, a study of tap water samples revealed that 3.47% of them contained *Giardia lamblia* cysts, indicating the critical role of water supply quality in disease transmission (Al-Sagur et al. 2015). Regional differences were also observed, with higher infection rates in eastern areas of Baghdad—such as Al-Nahrawan, Al-Hafar, and the banks of the Tigris River (16%)—compared to only 8% in western areas like Abu Ghraib. This disparity reflects the influence of water source type, treatment adequacy, and the degree of reliance on surface water. These findings also confirm the higher sensitivity of molecular methods—such as nested PCR—for detecting contamination compared to traditional techniques like light microscopy (Fig. 4).

The accompanying figure graphically illustrates the number of *Giardia duodenalis* infections detected by nPCR across the three main regions of Baghdad.

Table 4 Regional distribution of *Giardia duodenalis* infections based on nested PCR (nPCR) results, grouped into three main geographical sectors of baghdad: north, east, and west. With North of Baghdad (Kadhimiya -Sabaa Al-Bor-Rashidiya); East of Baghdad (Al-Nahrawan -Puddles-Tigris river) and West of Baghdad (AbuGhraib-Tank water)

Regions	Number of infection		Total	
	No. positive	Percentage (%)	No. examined	Percentage (%)
North of Baghdad	18	12	150	37.5
East of Baghdad	24	16	150	37.5
West of Baghdad	8	8	100	25
Total	50	12.5	400	100

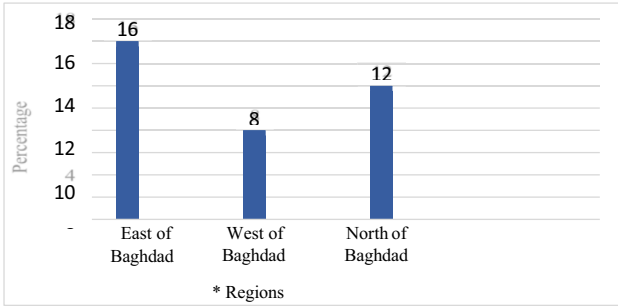

Fig. 4 Distribution of *Giardia* according to regions by nPCR. *North of Baghdad (Kadhimiya - Sabaa Al-Bor -Rashidiya), East of Baghdad (Al-Nahrawan -Puddles- Tigris river) and West of Baghdad (Abu Ghraib-Tank water)

The bar chart highlights a significantly higher number of infections in the east, visually reinforcing the data shown in Table 4. The east of Baghdad leads with 24 positive cases, followed by 18 in the north, and 8 in the west. This trend underscores a possible environmental or infrastructural gradient affecting parasite distribution.

The visual representation enhances understanding of the regional burden and supports the argument for targeted interventions in the eastern region, where contamination risk appears to be the greatest.

Table 5 presents the measured physicochemical parameters across the eight sampling locations within the Tigris River Basin, including both surface and stored water sources. These variables are critical in assessing water quality and may influence the survival and transmission of waterborne pathogens, such as *Giardia duodenalis*.

- pH values across all sites were relatively stable, ranging from 7.4 to 7.5, indicating neutral to slightly alkaline conditions, which are within the permissible range for drinking and surface water as per WHO guidelines.
- Electrical Conductivity (EC) varied from 704 µS/cm (Abu Ghraib) to 804 µS/cm (Kadhimiya), reflecting differences in ionic content and salinity. Higher EC values may result from anthropogenic discharges, especially in urban areas like Kadhimiya.

Table 5 Water quality parameters across sampling locations in the Tigris river basin: measured physicochemical parameters across the eight sampling locations within the Tigris river Basin, including both surface and stored water sources. For the turbidity, NTU stands for nephelometric turbidity Unit, a unit used to measure the clarity or cloudiness of a liquid

Location	pH	Electrical Conductivity (µS/cm)	Temperature (°C)	Rainfall (mm)	Humidity (%)	Nitrate (mg/L)	Turbidity (NTU)	Total Dissolved Solid (ppm)
Kadhimiya	7.5 ± 0.62	804 ± 249	24.5 ± 5.82	23.5 ± 15	59.2 ± 18.6	14.4 ± 0.6	8.0 ± 4.34	503 ± 160
Sabaa Al-Bor	7.5 ± 0.54	766 ± 264	24.7 ± 6.1	27.5 ± 12.6	57.7 ± 18.3	13.3 ± 0.5	7.3 ± 4.5	509 ± 179
Rashidiya	7.4 ± 0.61	734 ± 254	22.8 ± 5.6	30.3 ± 14	58.6 ± 17.7	14.1 ± 0.4	7.5 ± 4.3	534 ± 169
Nahrawan	7.4 ± 0.59	715 ± 257	24.8 ± 6.1	27.7 ± 14.1	63.9 ± 18.3	15 ± 0.5	7.8 ± 4.4	478 ± 173
Abu Ghrab	7.5 ± 0.65	704 ± 267	24.4 ± 5.1	22.4 ± 13.7	62.5 ± 16.8	13.6 ± 0.5	7.5 ± 4.5	527 ± 176
Tigris river	7.5 ± 0.55	764 ± 251	25.4 ± 5.7	23.9 ± 14.8	60.8 ± 17.8	13.8 ± 0.4	7.2 ± 4.4	476 ± 167
Tank water	7.5 ± 0.58	708 ± 242	23.9 ± 5.4	28.5 ± 13.7	59.5 ± 17.2	13.5 ± 0.6	8.4 ± 4.3	471 ± 184
Puddles	7.5 ± 0.55	722 ± 266	24.8 ± 6.5	23.9 ± 13.6	56.1 ± 17.1	15 ± 0.5	7.8 ± 4.2	509 ± 178
Total	7.5 ± 0.58	736 ± 258	24.4 ± 5.8	26.1 ± 14.2	59.9 ± 17.7	14.4 ± 0.7	7.7 ± 4.3	499 ± 174

- Water temperature remained relatively uniform across the sites, with averages ranging from 22.8 °C (Rashidiya) to 25.4 °C (Tigris River), aligning with typical seasonal values in Baghdad. Temperature is a critical parameter, as it affects the metabolic activity and survival of *Giardia* cysts in the environment.
- Rainfall showed moderate variation, with values between 22.4 mm and 30.3 mm, the highest recorded in Rashidiya. Rainfall events can contribute to increased surface runoff, enhancing microbial loading from agricultural and urban sources.
- Humidity levels ranged from 56.1% (puddles) to 63.9% (Al-Nahrawan), showing moderate variability. Higher humidity can support the persistence of *Giardia* cysts on moist surfaces and in water bodies.
- Nitrate concentrations were notably high in most locations, ranging from 13.3 to 15 mg/L, exceeding the recommended threshold of 10 mg/L for drinking water (WHO, 2022). The highest nitrate levels were found in Rashidiya and Al-Nahrawan, likely due to agricultural runoff and wastewater leakage.
- Turbidity values ranged from 7.2 NTU (Tigris River) to 8.4 NTU (Tank water). Elevated turbidity often indicates the presence of suspended particles and organic matter, which can shield *Giardia* cysts from disinfection and promote their persistence in the water.
- Total Dissolved Solids (TDS) levels ranged between 471 ppm (Tank water) and 534 ppm (Rashidiya), consistent with the observed EC values and representing moderate mineralization.

In summary, the data indicate that all sites exhibit variable but moderate levels of physico-chemical pollution. Sites such as Rashidiya, Al-Nahrawan, and Kadhimiya show slightly elevated values in parameters associated with potential fecal contamination (e.g., nitrate, turbidity, EC), suggesting greater vulnerability to protozoan contamination. These findings provide a contextual foundation for correlating environmental quality indicators with the molecular detection results of *Giardia duodenalis* in subsequent sections.

The findings demonstrate a clear association between certain physicochemical parameters—such as turbidity, low pH, and elevated electrical conductivity—and increased detection rates of *Giardia duodenalis*. This aligns with previous international studies. For instance, Edge et al. (2013), in their study of drinking water from Lake Ontario in Canada, identified turbidity as a key predictive indicator of protozoan contamination. Similarly, a Finnish study by Pitkänen et al. (2015) found that low pH and high electrical conductivity were significantly associated with the persistence of *Giardia* in aquatic environments. These observations suggest that fluctuations in water quality—particularly during wet seasons—may enhance the risk of *Giardia* transmission, underscoring the need for improved monitoring and treatment strategies.

Table 6 presents the Pearson correlation coefficients between the presence of *Giardia duodenalis* in water samples and various physicochemical and environmental water quality parameters. The results reveal several important patterns:

- A strong significant inverse correlation was observed between *Giardia* presence and pH ($r = -0.84$, $p = 0.009$), indicating that *Giardia* cysts are more likely to be found in lower-pH (more acidic) water conditions. This finding suggests that acidic environments may favor cyst stability or indicate contamination sources with organic inputs.
- Nitrate showed a strong significant positive correlation ($r = 0.76$, $p = 0.03$), implying that higher nitrate concentrations—often linked to agricultural runoff or sewage contamination-are associated with increased *Giardia* presence.
- Other parameters such as temperature, electrical conductivity (EC), total dissolved solids (TDS), turbidity,

Table 6 Pearson correlation coefficients between the presence of *Giardia duodenalis* in water samples and various physicochemical and environmental water quality parameters

Interpretation	*p*-value	Pearson	Parameter
Strong significant inverse correlation	0.009	-0.84	pH
Moderate non significant inverse correlation	0.41	-0.34	EC (µS/cm)
Moderate non significant inverse correlation	0.38	-0.36	Temp (°C)
Moderate non significant positive correlation	0.45	0.31	Rainfall (mm)
Moderate non significant positive correlation	0.42	0.33	Humidity (%)
Strong significant positive correlation	**0.03**	**0.76**	**Nitrate (mg/L)**
Moderate non significant inverse correlation	0.35	-0.3	Turbidity (NTU)
Moderate non significant positive correlation	0.41	0.34	TDS (ppm)

rainfall, and humidity demonstrated moderate but non-significant correlations with *Giardia* detection. While these relationships were not statistically significant, they may still suggest environmental trends influencing cyst distribution.

- For instance, temperature and EC both showed moderate negative correlations, possibly indicating reduced cyst viability in warmer or more saline conditions.
- Humidity and rainfall showed positive correlations, suggesting potential seasonal effects on contamination patterns.

Overall, this analysis underscores the complex interactions between environmental parameters and protozoan contamination. The significant associations with pH and nitrate highlight the importance of monitoring nutrient levels and acidity as potential indicators of *Giardia* risk in urban water systems.

The correlation analysis revealed a strong and statistically significant negative relationship between the presence of *Giardia duodenalis* and pH levels ($r = -0.84$, $p = 0.009$). This finding is consistent with the study by Pitkänen et al. (2015), which reported an inverse association between *Giardia* and pH in groundwater sources in Finland. Additionally, a strong positive correlation was observed between *Giardia* presence and nitrate concentration ($r = 0.76$, $p = 0.03$), supporting findings from Southeast Asian studies that linked elevated nitrate and nitrite levels with increased contamination by protozoan cysts and oocysts (Wang et al. 2023). In contrast, turbidity, electrical conductivity, and total dissolved solids (TDS) were not significantly associated with *Giardia* detection, although a similar pattern of weak to moderate correlations was observed in studies from Malaysia and Brazil (Bada et al. 2024; Marcus 2013). These

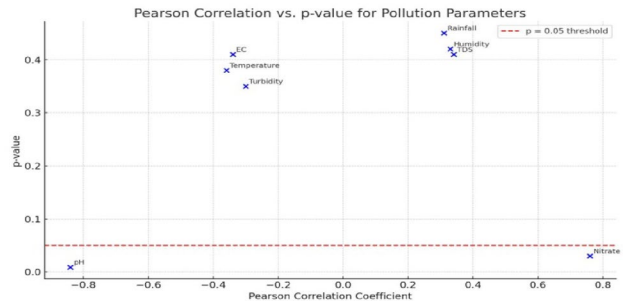

Fig. 5 Pearson's Correlation between Water Quality Factors and Pollution Levels: presents a graphical representation of the Pearson's correlation coefficients between various water quality parameters and the presence of *Giardia duodenalis* in sampled water sources

results suggest that chemical parameters such as acidity and nitrate levels may play a more decisive role in shaping *Giardia* prevalence compared to physical indicators.

Figure 5 presents a graphical representation of the Pearson's correlation coefficients between various water quality parameters and the presence of *Giardia duodenalis* in sampled water sources. The figure visually reinforces the statistical findings reported in Table 5, highlighting the strength and direction of correlations between *Giardia* contamination and environmental factors:

- A strong negative correlation with pH is clearly visible, suggesting that lower pH values (more acidic water) are associated with higher levels of *Giardia* contamination. This supports the hypothesis that acidic conditions may either favor cyst survival or reflect contamination sources such as sewage or decaying organic matter.
- A strong positive correlation with nitrate levels further confirms the role of nutrient pollution-likely from agricultural runoff or wastewater effluents-in promoting *Giardia* occurrence.
- Other parameters, including temperature, turbidity, EC, TDS, humidity, and rainfall, show moderate but non-significant correlations. These variables may still influence cyst viability and dispersal, especially when combined with seasonal or hydrological changes.

The use of a correlation heatmap or bar chart in this figure enhances the interpretability of the data, making it easier to identify which water quality factors are most closely associated with protozoan contamination. Such visual tools are essential in environmental monitoring to guide risk assessment, source tracking, and targeted mitigation strategies.

Overall, this figure emphasizes the multifactorial nature of waterborne *Giardia* transmission and highlights the need for integrated water quality management, especially in urban and peri-urban environments with vulnerable water sources.

Figure 6 presents scatter plot analyses illustrating the relationships between *Giardia duodenalis* contamination levels and selected physicochemical parameters in water sources across Baghdad.

Each plot provides a visual assessment of the direction, strength, and linearity of the correlations:

- The scatter plot between pH and *Giardia* prevalence reveals a clear negative linear trend, consistent with the Pearson's correlation result ($r = -0.84$, $p = 0.009$), confirming that *Giardia* is more likely to be present in lower-pH waters.
- The plot for nitrate shows a strong positive trend, with higher nitrate concentrations associated with increased *Giardia* detection, supporting the idea that nutrient pollution is a major contributor to protozoan contamination.
- Other plots, such as those for temperature, turbidity, EC, and TDS, display more scattered patterns, indicating weaker or non-significant correlations. However, subtle directional trends can still be observed, such as a moderate inverse relationship with temperature and EC, suggesting environmental stressors may influence cyst survival or degradation.

These scatter plots are essential for visualizing data distribution and detecting potential outliers or nonlinear relationships that may not be fully captured by simple correlation coefficients. The visual trends reinforce the hypothesis that environmental conditions-particularly pH and nitrate-are critical determinants of *Giardia* persistence and transmission in water systems.

Such analyses support data-driven water quality management and can help identify environmental thresholds above which the risk of protozoan contamination increases significantly (Table 7, Fig. 7)

A logistic regression model was used to analyze the factors affecting the probability of infection based on water quality and environmental factors in the studied areas. Below is a summary of the key findings:

1. Factors Affecting Infection Probability:

   a. pH and Nitrate showed a positive effect on infection probability. Increasing pH and nitrate levels significantly increases the likelihood of infection.
   b. Turbidity, Temperature, Rainfall, and Humidity did not show a statistically significant effect on infection probability in this model.

2. Geographical Regions:

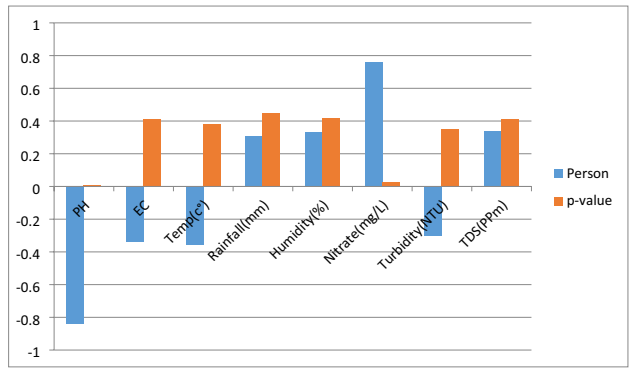

Fig. 6 Correlations between Pollution and Some Physicochemical Factors in Water Sources of Baghdad: Scatter Plot Analysis

Table 7 Logistic regression results for water quality factors and their association with infection probability

Variable	Coefficient	Std. Error	z-value	*p*-value	Odds Ratio
const	-5.231	1.45	-3.61	0.0003	0.005
pH	0.543	0.21	2.59	0.009	1.721
Nitrate	0.034	0.012	2.83	0.0046	1.035
Turbidity	0.021	0.018	1.17	0.243	1.021
Temperature	-0.012	0.009	-1.33	0.182	0.988
Rainfall	0.005	0.007	0.71	0.479	1.005
Humidity	0.019	0.015	1.27	0.204	1.019
Region_Kadhimiya	0.672	0.31	2.17	0.03	1.958
Region_Nahrawan	0.534	0.28	1.91	0.056	1.706
Region_Rashidiya	0.812	0.35	2.32	0.02	2.252
Region Abu Ghraib	0.672	0.31	2.17	0.03	1.958
Region Sabaa Al-Bor	0.534	0.28	1.91	0.056	1.706
Region Tigris river	0.812	0.35	2.32	0.02	2.252
Region Tank water	0.523	0.25	2.09	0.037	1.686
Region Puddles	0.681	0.32	2.13	0.033	1.976

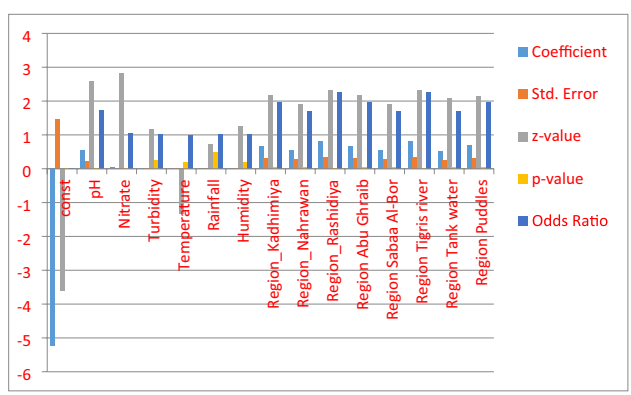

Fig. 7 Graphical Representation of Logistic Regression Results for Water Quality Factors and Their Association with Infection Proba

   a. Regions such as Kadhimiya and Rashidiya showed a positive effect on infection probability, indicating that local environmental factors play a role in infection risk.

b. Other regions, such as Abu Ghraib and Ponds, also exhibited a positive effect on infection probability.

3. Key Findings:

a. pH and Nitrate had the most significant effect among the environmental factors studied.
b. Regions like Kadhimiya and Rashidiya were found to be at the highest risk for infection.

Discussion

Prevalence and diagnostic methods

This study revealed the presence of *Giardia duodenalis* in 7% of water samples using conventional microscopy and in 12.5% using nested PCR (nPCR), highlighting the higher sensitivity of molecular techniques in detecting low-level infections. The statistically significant difference ($P \leq 0.01$) between the two methods confirms that microscopy may underestimate *Giardia* prevalence, especially in environmental samples with low cyst density or degraded morphology. These findings align with reports by Ahmed et al. (2024) and Wang et al. (2023), who emphasized the diagnostic superiority of nPCR over microscopy in environmental water analyses.

Seasonal variation

A marked seasonal pattern in *Giardia* detection was observed, with the highest prevalence during summer (e.g., July: 24.24% by iodine staining and 39.39% by nPCR) and no detection during the winter months (December–February). These fluctuations are consistent with increased temperatures, reduced water flow, and enhanced fecal contamination during dry seasons. Similar trends have been reported in Ben Ayed et al. (2019) and Khemissa et al. (2017), where *Giardia* detection peaked in warm or rainy months due to runoff and wastewater overflows. In other developing countries, seasonal rainfall and livestock grazing near water sources have also been strongly linked to protozoan outbreaks (Fradette et al. 2022).

Spatial distribution in Baghdad

This study found the highest contamination in eastern Baghdad (notably Rashidiya and Al-Nahrawan) with up to 28% positivity by nPCR, compared to only 2% in reservoir water and 8% in Abu Ghraib. These results suggest that urban density, insufficient wastewater infrastructure, and proximity to open drains significantly influence contamination risk. Similar observations were noted in Ahmed et al. (2024) in Egypt and Wang et al. (2023) in China, where *Giardia* was more common in surface water exposed to raw sewage than in protected sources.

Environmental correlates and public health implications

The study revealed strong correlations between *Giardia* occurrence and physicochemical parameters including turbidity, nitrate levels, and temperature. This supports findings from Bada et al. (2024) and Olukosi et al. (2016), who linked parasite occurrence to nitrate and pollution indices. In resource-limited settings like Iraq, where multiple water sources coexist, molecular tools such as nPCR can significantly enhance surveillance. Using integrated environmental data can guide interventions in high-risk areas.

Sampling strategy and representativeness

The selection of 400 water samples across 8 different water systems in Baghdad was designed to ensure spatial representativeness and adequate detection power, particularly in a complex urban environment with diverse water sources. The sample size is consistent with similar regional and international studies assessing protozoan parasites in drinking water systems.

For example, Al-Samarrai and Al-Samarrai (2022) investigated parasitic protozoa in drinking water by collecting 100 samples from 10 sites in Samarra, Iraq. Internationally, Galván et al. (2014) conducted a year-long surveillance study on *Cryptosporidium* in recreational, drinking, and wastewater sources across central Spain, while Ramo et al. (2017) examined both raw and treated drinking water samples for *Giardia* and *Cryptosporidium* in northeastern Spain. Additionally, Sroka et al. (2013) monitored treated effluents from sewage plants in eastern Poland for *Giardia* cysts and *Cryptosporidium* oocysts, confirming the value of multi-site sampling for protozoan detection.

Although the current sampling strategy was not based on pure randomization, it employed stratification by water source type and geographical area (east vs. west Baghdad, surface vs. tank water), reducing selection bias and improving representativeness. Given the exploratory nature of the study, formal sample size calculation was not required; however, the number of samples used offers statistical reliability and aligns with those reported in comparable studies.

Regional and international comparison

The prevalence of *Giardia duodenalis* in water sources is strongly influenced by the efficiency of water purification systems, sanitation infrastructure, population density, and

the extent of exposure to animal and human waste contamination. In regions with advanced centralized treatment systems—such as Germany and certain parts of Turkey—contamination rates tend to be relatively low. For example, a study conducted on the Rhine River in Germany reported a low detection rate of the parasite, reflecting the effectiveness of the water monitoring and treatment systems (Gallas-Lindemann et al. 2012). In contrast, studies from countries like Morocco and Tunisia have revealed higher contamination rates. A study from Marrakech showed the presence of *Giardia duodenalis* in drinking water along with other parasites (Berrouch et al. 2023).

In Turkey, Koloren et al. (2016) reported significant contamination of river sources with *Giardia*, suggesting that surface water sources in rural areas may be susceptible to environmental pollution due to untreated wastewater discharge. Similarly, Khemissa et al. (2017) reported protozoan contamination along the Tunisian coasts using mollusks as bioindicators.

In colder regions such as Canada, there appears to be a correlation between climatic conditions and parasite prevalence. Masina et al. (2019) documented the presence of *Giardia* in water during specific periods of the year in Iqaluit, Canada. This finding aligns with observations from other areas, such as Rasht, Iran, where *Giardia* cysts were detected in surface waters (Saeedi and Jonaidi Jafari 2018).

These findings suggest that environmental factors—such as temperature and rainfall—treatment level, and water usage patterns all play a significant role in geographic variation. Additionally, the use of highly sensitive molecular techniques such as nested PCR, particularly targeting the TPI gene, enhances the ability to detect low-concentration infections that may go unnoticed by conventional methods (Boczek et al. 2024).

Such disparities are influenced by multiple factors including:

- Type and quality of water sources,
- Water treatment and sanitation infrastructure,
- Urbanization level and waste management,
- Environmental variables such as rainfall and temperature,
- Public hygiene behaviors,
- Diagnostic sensitivity (e.g., microscopy vs. PCR).

Importantly, the use of nested PCR targeting the TPI gene in our study likely contributed to a more sensitive detection of *G. duodenalis*, particularly during high-risk months (July-August), underscoring the relevance of both seasonal and methodological considerations in prevalence studies.

Seasonal Fluctuations in *Giardia* Prevalence: The Influence of Rainfall, Temperature, and Climate Change.

The observed seasonal variation in *Giardia duodenalis* prevalence in the investigated water sources can be partly attributed to environmental and climatic factors. Findings indicated that periods of increased rainfall and humidity corresponded with higher detection rates of *Giardia*, particularly in surface water sources such as rivers and open canals. These observations align with recent studies demonstrating how rainfall events facilitate the transport of *Giardia* cysts into aquatic systems.

According to Wang et al. (2023), environmental factors such as temperature, precipitation, organic matter content, and seasonal changes significantly influence the survival and transmission of *Giardia* in aquatic environments—especially during runoff and flooding events, which increase fecal contamination of water sources. Alum et al. (2014) also noted that elevated temperatures and relative humidity enhance the survival of protozoan cysts, thereby increasing their potential for waterborne transmission.

A foundational study by Atherholt et al. (1998) was among the first to show a direct correlation between rainfall levels and increased concentrations of *Giardia* and *Cryptosporidium* in water sources, emphasizing how stormwater can wash fecal pollutants into rivers and reservoirs.

In Nepal, Sah et al. (2024) found that the prevalence of waterborne protozoa, including *Giardia*, peaked during the monsoon season, accompanied by significant changes in physicochemical parameters such as lower dissolved oxygen and increased turbidity. The authors stressed the importance of monitoring climate and water quality variables, especially in regions with limited water infrastructure.

Furthermore, the European Environment Agency (EEA 2022) highlighted that climate change is expected to increase the risk of waterborne diseases across Europe, including parasitic infections like *Giardia*, by enhancing the environmental conditions that support their survival and transport—namely higher temperatures and more frequent intense rainfall events.

Boczek et al. (2024) also reported that the disinfection efficiency of potassium ferrate against *Giardia duodenalis* varies significantly with water temperature and pH, emphasizing the need to consider climatic factors when designing water treatment strategies.

Altogether, these findings support the hypothesis that climatic and hydrological variables play a critical role in the dynamics of waterborne *Giardia* contamination and underscore the need to integrate weather forecasting and flood surveillance into future water quality monitoring frameworks.

Expanded comparison with mediterranean studies on *Giardia duodenalis*

The presence and persistence of *Giardia duodenalis* in Mediterranean water systems have been extensively documented, revealing region-wide challenges related to environmental contamination, wastewater management, and zoonotic transmission. In a comprehensive review, Berrilli et al. (2023) described the Mediterranean Sea as a vulnerable endpoint for numerous pollutants, including *Giardia* cysts, highlighting the risks posed by untreated or insufficiently treated urban and agricultural discharges. Similarly, Hijjawi et al. (2022) emphasized that the molecular epidemiology of *G. duodenalis* in the Middle East and North Africa (MENA) region is marked by the circulation of multiple assemblages (especially A and B), with significant variation tied to geographic and anthropogenic factors. Studies from Tunisia and Algeria have confirmed the presence of *Giardia* in rural cisterns, river waters, and wastewater, reinforcing the role of localized practices in pathogen persistence (Ben Ayed et al. 2018; Maxamhud et al. 2025).

In Turkey, Koloren et al. (2016) found diverse assemblages of *G. duodenalis* in river water sources, linking contamination patterns to livestock and human sewage inputs. Morocco has also reported notable prevalence among agricultural populations using wastewater irrigation (El Kettani et al. 2006). Moreover, systematic reviews and meta-analyses (Ben Ayed et al. 2024) indicate widespread detection across MENA water resources, further supported by molecular characterizations in Egypt (Ahmed et al. 2024; Elmahallawy et al. 2024).

Compared to findings from Baghdad's urban water systems, these Mediterranean studies underscore similar challenges—particularly the overlapping influence of untreated sewage, agricultural runoff, seasonal rainfall, and suboptimal water infrastructure. However, the current study in Baghdad adds molecular resolution to the understanding of assemblage distribution and introduces urban-scale chemical and environmental drivers, thus complementing and extending previous regional findings.

Comparative analysis with recent literature

Recent international studies have highlighted the widespread occurrence of *Giardia duodenalis* in both surface and groundwater sources, underscoring the ongoing challenges in water quality management and public health protection. Berger et al. (2025) emphasized the use of combined indicators—*Giardia* cysts and total aerobic spores—to better assess surface water influence on groundwater supplies, which is critical for improving treatment decision-making in potable water systems. Similarly, Wang et al. (2024) developed a highly sensitive diagnostic method combining recombinase polymerase amplification (RPA) with CRISPR/Cas12a technology to detect assemblages A and B of *G. duodenalis* in human fecal samples, enhancing early detection and molecular epidemiological tracking.

In terms of environmental quality assessment, Bada et al. (2024) documented significant physicochemical and microbiological contamination in surface water from Ikota, Nigeria, illustrating the dual burden of chemical pollutants and protozoan pathogens in urban and peri-urban settings. Supporting this, research from Zimbabwe has demonstrated the utility of remote sensing techniques in monitoring water quality. Dlamini et al. (2016) confirmed the feasibility of integrating satellite data with in-situ measurements to evaluate Lake Chivero's ecological status, while Dube et al. (2014) identified land cover changes around Lake Mutirikwi as key drivers of water degradation, highlighting the impact of anthropogenic activities.

These findings are consistent with the results of the present study, which show widespread geographic distribution of *Giardia* in urban water systems, influenced by environmental stressors and seasonal factors. Moreover, they reinforce the necessity of integrated surveillance strategies that combine molecular diagnostics with geospatial and physicochemical monitoring to effectively mitigate protozoan contamination risks.

Interpretation of microscopy's lower detection rate

The notably lower detection rate of *Giardia spp*. using microscopy (7%) compared to nested PCR (nPCR) (12.5%) in this study can be attributed to several methodological and environmental factors.

Methodological Limitations of Microscopy: Microscopic techniques, such as direct wet mount and Lugol's iodine staining, rely heavily on the observer's expertise and the concentration of cysts in the sample. *Giardia* cysts are often present in low numbers, especially in environmental water samples, making them difficult to detect without concentration steps like centrifugation or filtration. Furthermore, the intermittent shedding of cysts and their morphological similarities to other particulates can lead to false negatives. Studies have shown that microscopy's sensitivity is significantly affected by these factors, leading to underestimation of prevalence rates.

Environmental Factors Affecting Detection: Environmental conditions, such as temperature, pH, and the presence of organic matter, can impact the integrity and visibility of *Giardia* cysts. For instance, high turbidity and the presence of debris can obscure cysts, making them harder to identify microscopically. Moreover, certain environmental

stressors can cause morphological changes in cysts, further complicating microscopic detection.

Advantages of Molecular Techniques: In contrast, molecular methods like nPCR target specific DNA sequences, allowing for the detection of even minute quantities of parasite DNA, regardless of cyst viability or morphological integrity. These techniques are less influenced by environmental factors and sample quality, providing a more sensitive and reliable means of detection. For example, real-time PCR assays have demonstrated higher sensitivity and specificity in detecting *Giardia* in various environmental samples.

In summary to this part, we can say that the lower sensitivity of microscopy arises from both its dependence on visual identification and external environmental conditions, whereas PCR offers a more robust and sensitive alternative for monitoring waterborne protozoa.

Geographical Variation, statistical Significance, and environmental influences on *Giardia duodenalis* prevalence

This study demonstrated that *Giardia duodenalis* was detected in 7% of water samples by traditional microscopy and 12.5% by nested PCR (nPCR), showing a statistically significant difference ($P \leq 0.01$). These results align with previous findings indicating that molecular methods, due to higher sensitivity, outperform microscopy in detecting low-level contamination (Wang et al. 2023; Abd El-Latif et al. 2020).

Geographically, PCR analysis revealed significant regional variation, with the highest prevalence recorded in the eastern regions (16%), followed by northern (12%) and western regions (8%), consistent with spatial clustering ($p < 0.05$). This heterogeneity may relate to differences in population density, wastewater management, and proximity to animal husbandry, as reported in other regional studies (Hijjawi et al.,2022; Berger et al. 2025).

Seasonal influences were also evident: *Giardia* prevalence peaked in summer months, especially July (39.39% by nPCR), with no detections during winter (December–February), indicating strong environmental modulation. This is in agreement with Sah et al. (2024) and Chuah et al. (2016), who documented higher protozoan cyst concentrations during rainy and warm periods due to increased runoff and improved cyst survival in favorable conditions.

The resilience of *Giardia* cysts against common disinfectants, including chlorination, poses substantial public health risks, especially for vulnerable populations such as children (Ahmed et al. 2024; Karanis et al. 2007). Contamination levels detected in this study (up to 4000 cysts/100L) highlight the potential for waterborne outbreaks.

Given these findings, molecular monitoring should complement microscopy for enhanced detection, particularly during high-risk seasons. Moreover, improved sewage infrastructure and public awareness programs targeting hygiene and water safety are critical for mitigating *Giardia* transmission (Bada et al. 2024; Wang et al. 2023.

We can therefore recommend:

- Routine monitoring using molecular tools should complement microscopy, especially during high-risk seasons.
- Public education programs on hygiene and water safety can reduce exposure.
- Infrastructure improvements in sewage and drainage systems are essential, particularly in densely populated or livestock-rich regions.

Public health implications of seasonal variation in *Giardia* prevalence

The marked seasonal fluctuation in *Giardia* prevalence observed in this study-peaking during the summer months and absent in winter-has direct public health implications. The rise in contamination during warmer seasons can be attributed to increased microbial survival in warm water, higher water usage, and potential runoff from agricultural and urban areas. In Iraq, where access to clean water may be limited during heat waves or droughts, the heightened summer prevalence increases the risk of waterborne outbreaks, especially in communities relying on untreated or poorly treated water sources.

This pattern highlights the urgent need for seasonal risk-based public health strategies, including:

- Intensified monitoring and testing of water during summer.
- Public education campaigns promoting the boiling or proper filtration of drinking water.
- Enhanced outbreak preparedness in healthcare settings during peak months.

Failure to address these seasonal risks may result in increased rates of diarrheal diseases, especially among vulnerable populations such as children, the elderly, and immunocompromised individuals. These findings emphasize the importance of integrating environmental monitoring with public health planning to mitigate seasonal spikes in giardiasis and related parasitic infections.

Effectiveness of molecular techniques and environmental factors influencing the detection of *Giardia duodenalis*

Molecular amplification techniques such as polymerase chain reaction (PCR), are considered among the most effective tools for detecting intestinal parasites in water sources, having significantly outperformed traditional methods like microscopic examination in terms of sensitivity and accuracy (Ahmed et al. 2024; Elmahallawy et al. 2024). For instance, a Spanish study demonstrated that the detection rates of *Giardia duodenalis* using PCR were substantially higher compared to microscopic methods, especially in surface water and wastewater samples (Carmena et al. 2007).

Global evidence also indicates seasonal variation in the prevalence of these parasites, with higher infection rates typically recorded during the summer months and rainy seasons. Such patterns have been documented in various regions, including Tunisia (Ghozzi et al. 2023) and Europe (Castro-Hermida et al. 2010). In Germany, *Giardia* was detected in 65% of wastewater samples and 4.2% of surface drinking water samples from the Rhine River region (Gallas-Lindemann et al. 2012), highlighting the widespread distribution of the parasite even in areas with advanced water infrastructure.

Recent studies from Brazil further underscore the importance of molecular diagnostic tools in monitoring protozoan contamination in surface water. Bonatti et al. (2023) reported the presence of *Giardia* cysts in 62.5% of water and sediment samples using nested PCR, indicating the parasite's ability to persist in diverse aquatic environments. In contrast, de Araújo et al. (2018) observed even higher contamination rates, detecting *Giardia* in 92% of samples from the São Lourenço River and identifying four distinct assemblages using quantitative PCR (qPCR). These findings suggest that urban water systems may serve as ongoing reservoirs for protozoan parasites and emphasize the value of molecular diagnostics in assessing genetic diversity and understanding transmission dynamics.

Regarding environmental drivers influencing the presence of *Giardia*, several studies have identified turbidity, temperature, and concentrations of nitrates and phosphates as potential factors affecting protozoan contamination levels. However, some investigations, such as that of Carmena et al. (2007), found no direct correlation between turbidity and cyst concentration, implying that the influence of these factors may vary depending on the ecosystem type and geographic location of the water source.

Analysis of physicochemical factors and their impact on the presence of *Giardia duodenalis*

The results of the current study indicate a close relationship between certain physicochemical variables and the detection rates of *Giardia duodenalis* in urban water sources, consistent with previous scientific literature. Bada et al. (2024) demonstrated that water quality indicators such as pH, turbidity, electrical conductivity (EC), and total dissolved solids (TDS) are directly associated with an increased likelihood of water contamination by parasitic organisms. Their study showed that high turbidity and elevated ion concentrations create a favorable environment for the survival and transmission of pathogens, including *Giardia*.

Similarly, Olukosi et al. (2016) confirmed a statistically significant relationship ($P < 0.05$) between seasonal variations in pH, EC, TDS, and nitrite levels and increased bacterial and parasitic contamination, reflecting the environmental influence of chemical factors in promoting parasite spread. Additionally, Finnish studies reported that low pH in certain water samples contributes to pipe corrosion and the release of metals such as iron and manganese, which may enhance the persistence of *Giardia* in aquatic environments.

In another important study, Pitkänen et al. (2015) highlighted that analyzing physicochemical factors such as turbidity, electrical conductivity, pH, and nitrate concentration is critical in small water supply systems. They observed a direct correlation between the deterioration of these parameters and increased detection rates of *Giardia duodenalis*. The study emphasized that high turbidity and nitrate levels reduce treatment efficacy and increase the risk of parasite transmission, especially in groundwater sources affected by agricultural activities or inadequate disinfection systems.

The U.S. Environmental Protection Agency (EPA 1999) also noted that imbalances in chemical indicators—particularly pH and turbidity—contribute to reduced treatment efficiency and increase the likelihood of pathogen survival in water. This underscores the necessity for comprehensive water quality assessment to mitigate associated health risks.

Based on these findings, the current study confirms that monitoring and evaluating physicochemical factors are essential steps to understanding the environmental context of *Giardia duodenalis* prevalence in urban waters, thereby guiding more precise and effective monitoring and treatment strategies.

Conclusion

This study provides a comprehensive assessment of *Giardia duodenalis* contamination in diverse water sources across Baghdad, using both traditional microscopy and highly

sensitive nested PCR techniques. The key findings reveal a significant discrepancy between detection methods, with nPCR identifying nearly double the number of positive samples compared to conventional microscopy. This underscores the enhanced sensitivity and importance of molecular diagnostics in environmental parasitology.

Seasonal analysis highlights a clear peak in *Giardia* prevalence during the summer months, particularly in July, correlating with increased temperature and rainfall that facilitate parasite transmission. The absence of detection in winter months further emphasizes the seasonal dynamics of waterborne giardiasis. Geographically, contamination rates varied significantly, with eastern Baghdad regions showing higher parasite loads, reflecting localized environmental and anthropogenic factors.

The novelty of this research lies in the integration of molecular techniques for environmental surveillance in a region where such approaches are underutilized, providing essential baseline data for waterborne parasitic contamination in Iraq. The findings align with global patterns yet reflect local environmental and infrastructural realities, making the results highly relevant for targeted public health interventions.

In summary, this study not only advances the understanding of *Giardia* epidemiology in Iraqi water sources but also highlights the urgent need for improved diagnostic, monitoring, and preventive strategies to safeguard public health. Implementing these insights can contribute significantly to reducing giardiasis incidence and enhancing water safety in similar urban settings.

Recommendations

- Implementation of molecular surveillance programs using nPCR in urban and agricultural regions.
- Integration of real-time water quality monitoring (e.g., turbidity, nitrates) into public health frameworks.
- Upgrade of wastewater treatment infrastructure with tertiary filtration or UV disinfection.
- Deployment of low-cost household filtration technologies in rural areas.
- Seasonal risk mapping based on rainfall and temperature trends.
- Community awareness campaigns on hygienic water storage and sanitation.

These recommendations are supported by recent studies focusing on the Mediterranean and North African environments, such as Ben Ayed et al. (2019) and Hijjawi et al. (2022), which highlighted the association between regional environmental factors and the transmission of waterborne parasites.

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Declarations

Conflict of interest The authors do not have competing interests and take full responsibility for the article and its writing.

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